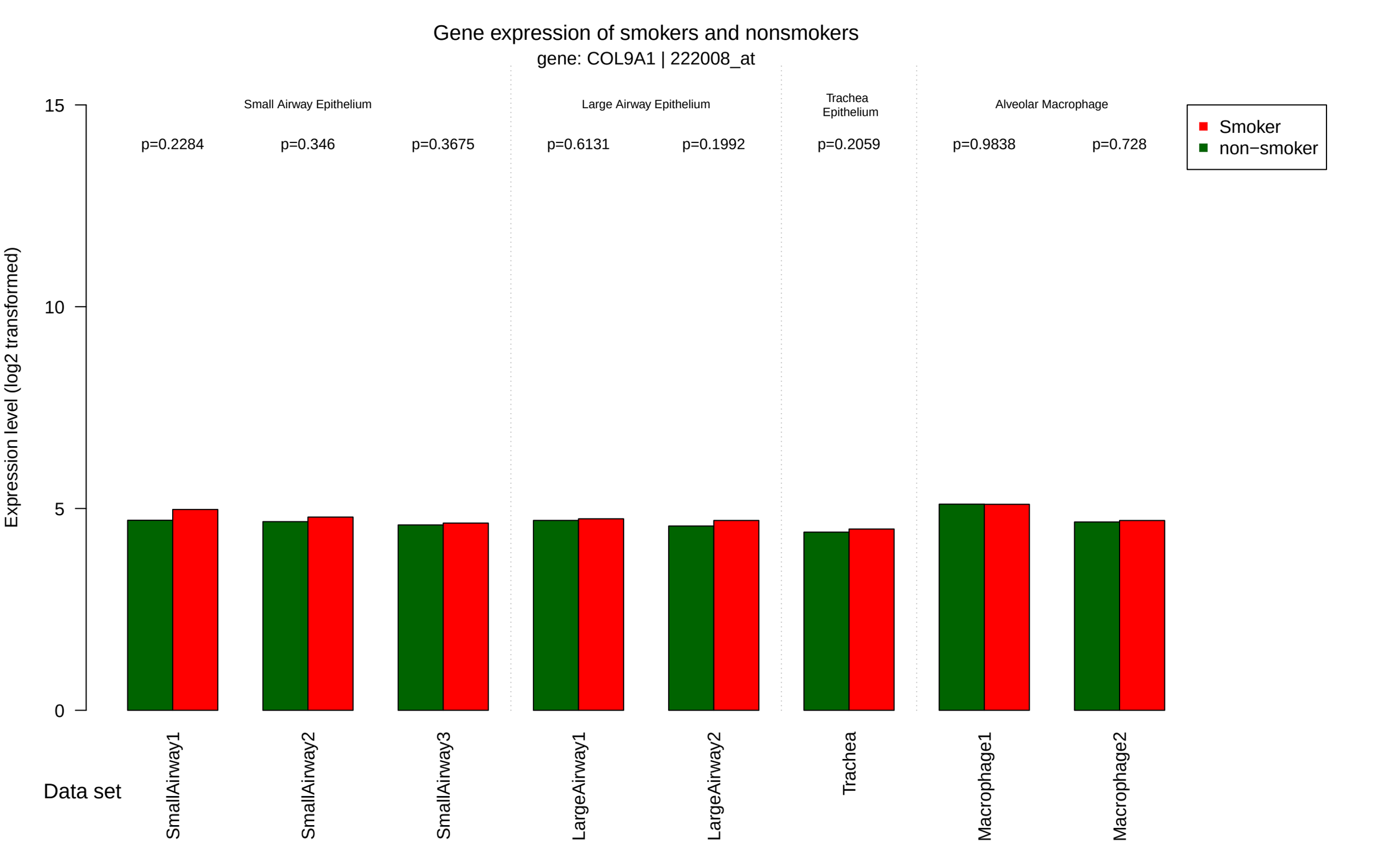
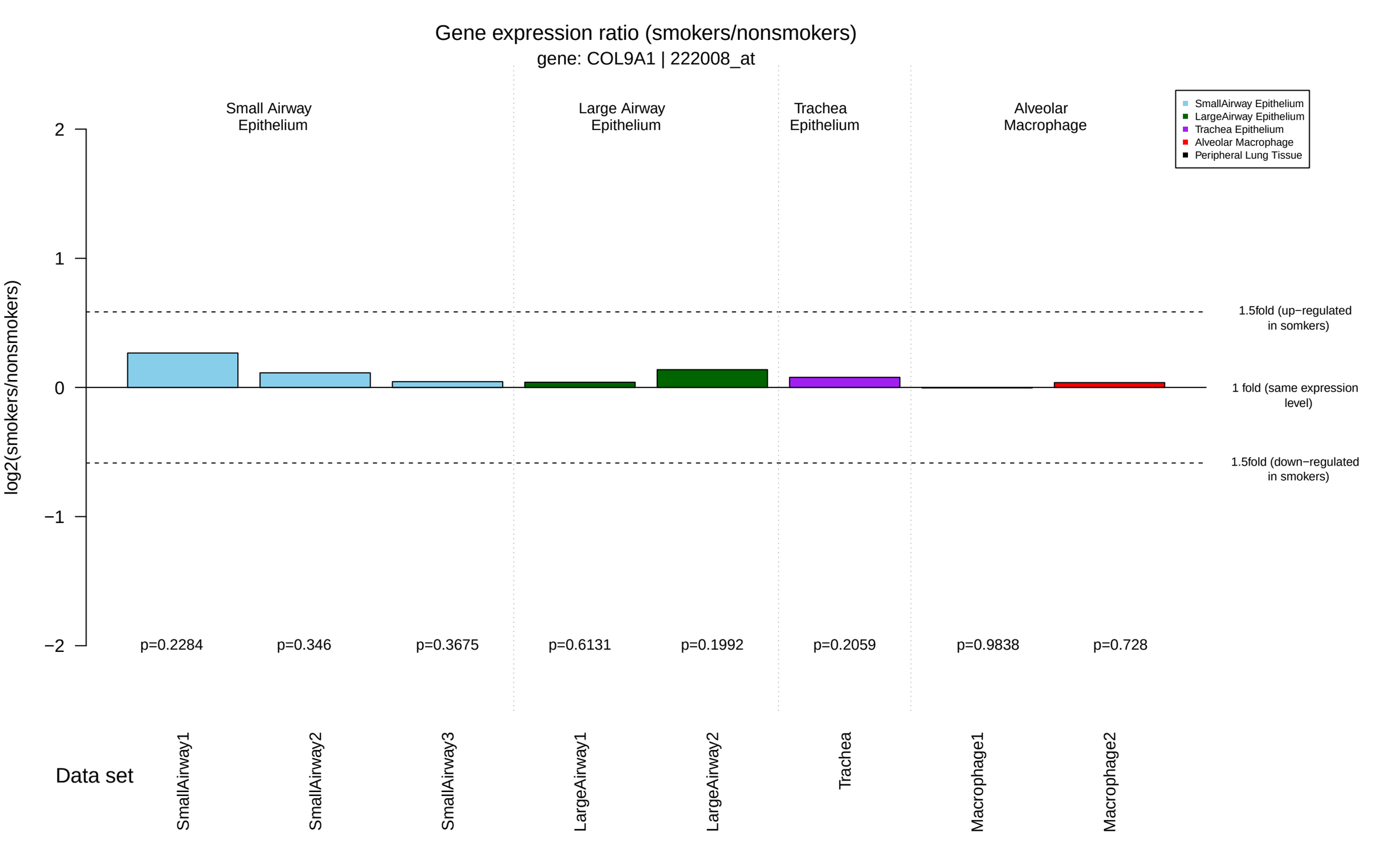
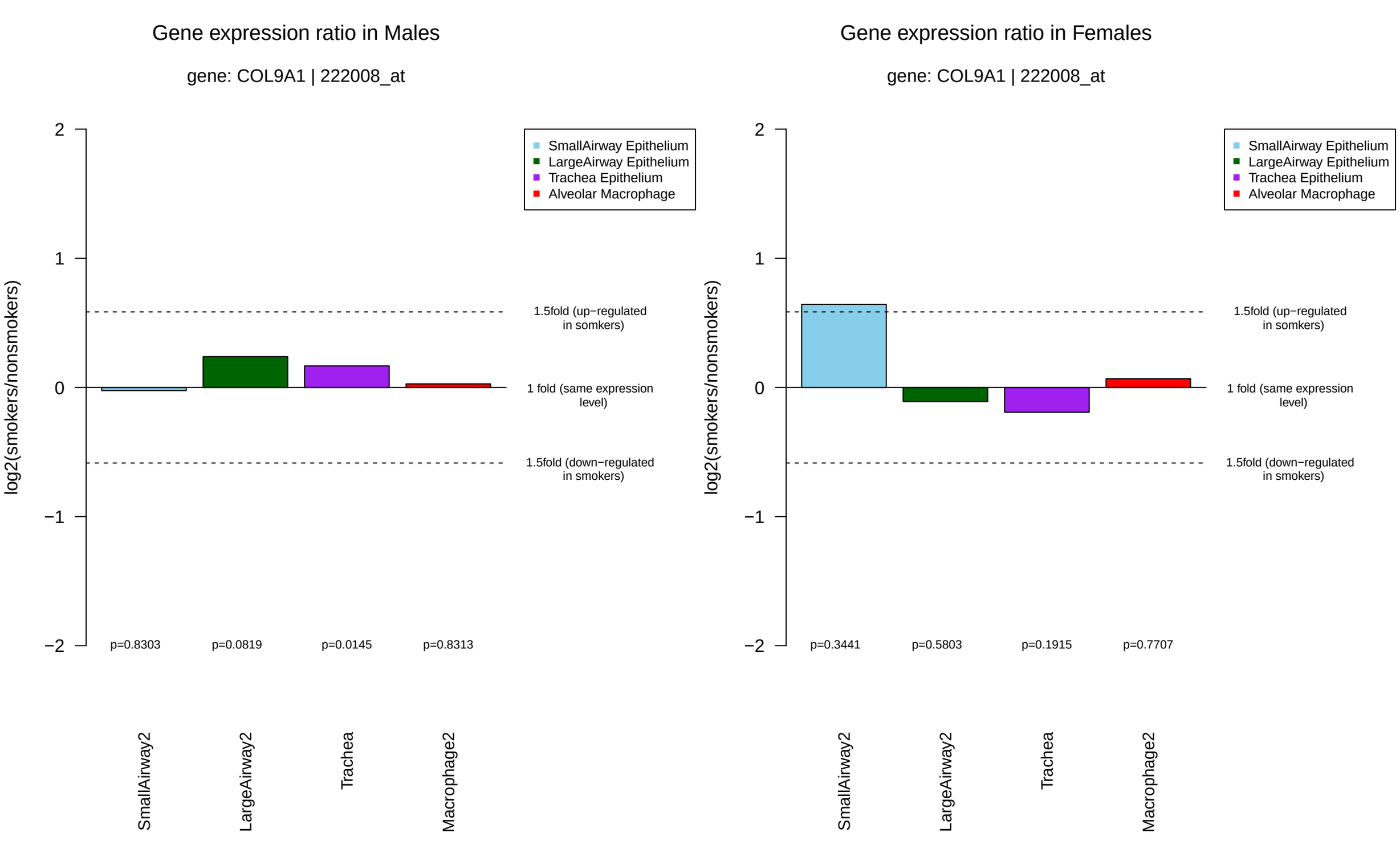
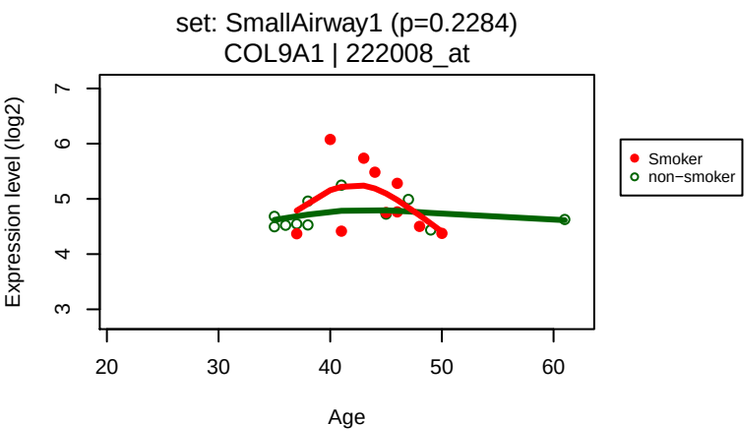
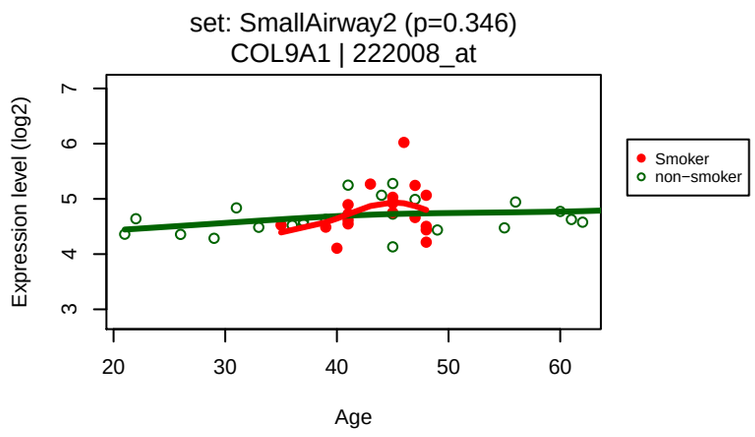
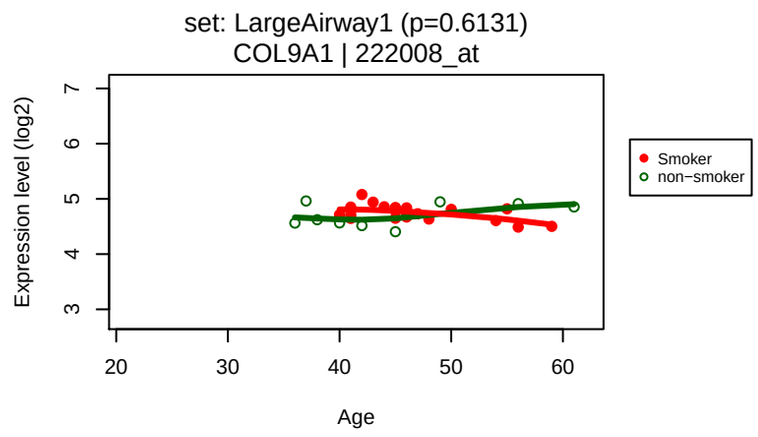

Gene expression of smokers and nonsmokers
gene: COL9A1 | 222008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2284
p=0.346
p=0.3675
p=0.6131
p=0.1992
p=0.2059
p=0.9838
p=0.728
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

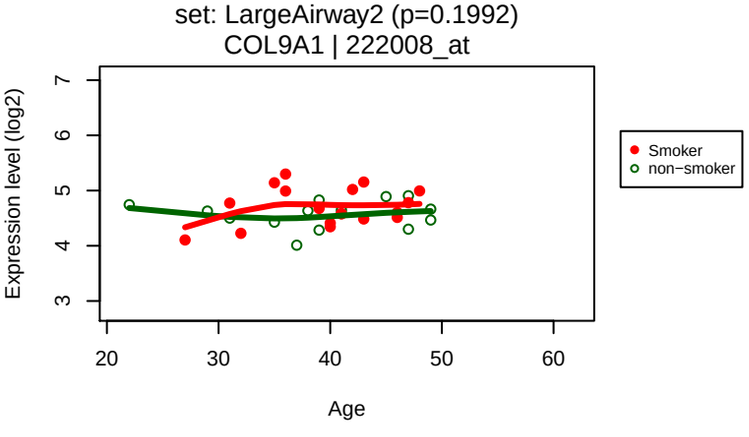

Gene expression ratio (smokers/nonsmokers)
gene: COL9A1 | 222008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2284
p=0.346
p=0.3675
p=0.6131
p=0.1992
p=0.2059
p=0.9838
p=0.728
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

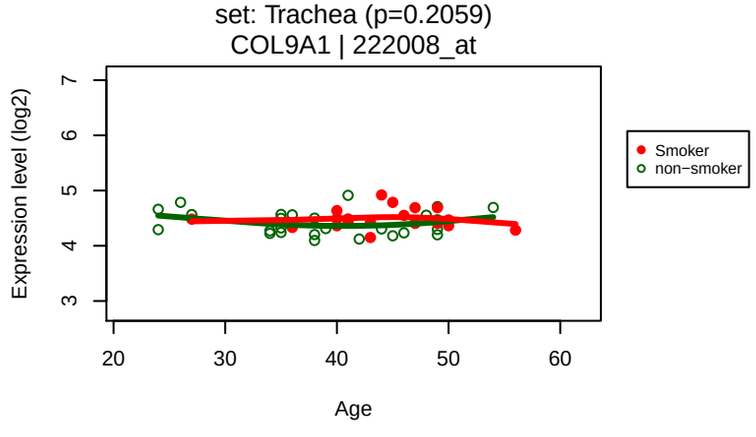

Gene expression ratio in Males
gene: COL9A1 | 222008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8303
p=0.0819
p=0.0145
p=0.8313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL9A1 | 222008_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3441
p=0.5803
p=0.1915
p=0.7707
SmallAirway2
LargeAirway2
Trachea
Macrophage2

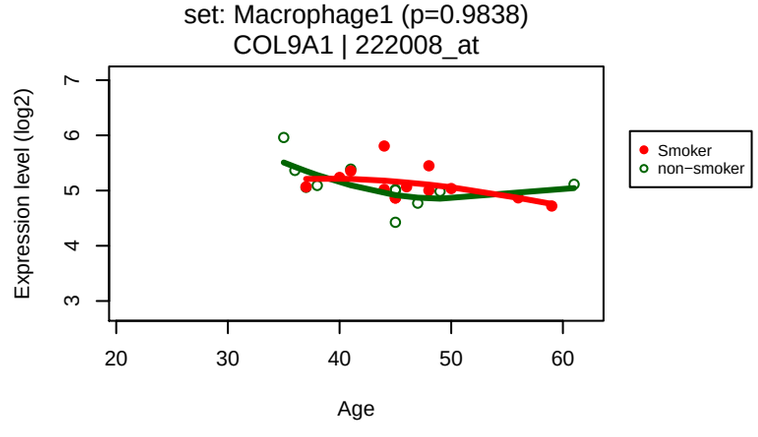

set: SmallAirway1 (p=0.2284)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker

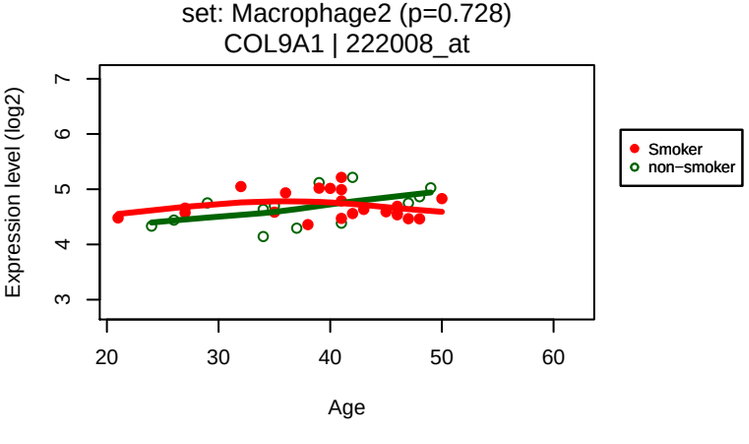

set: SmallAirway2 (p=0.346)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker

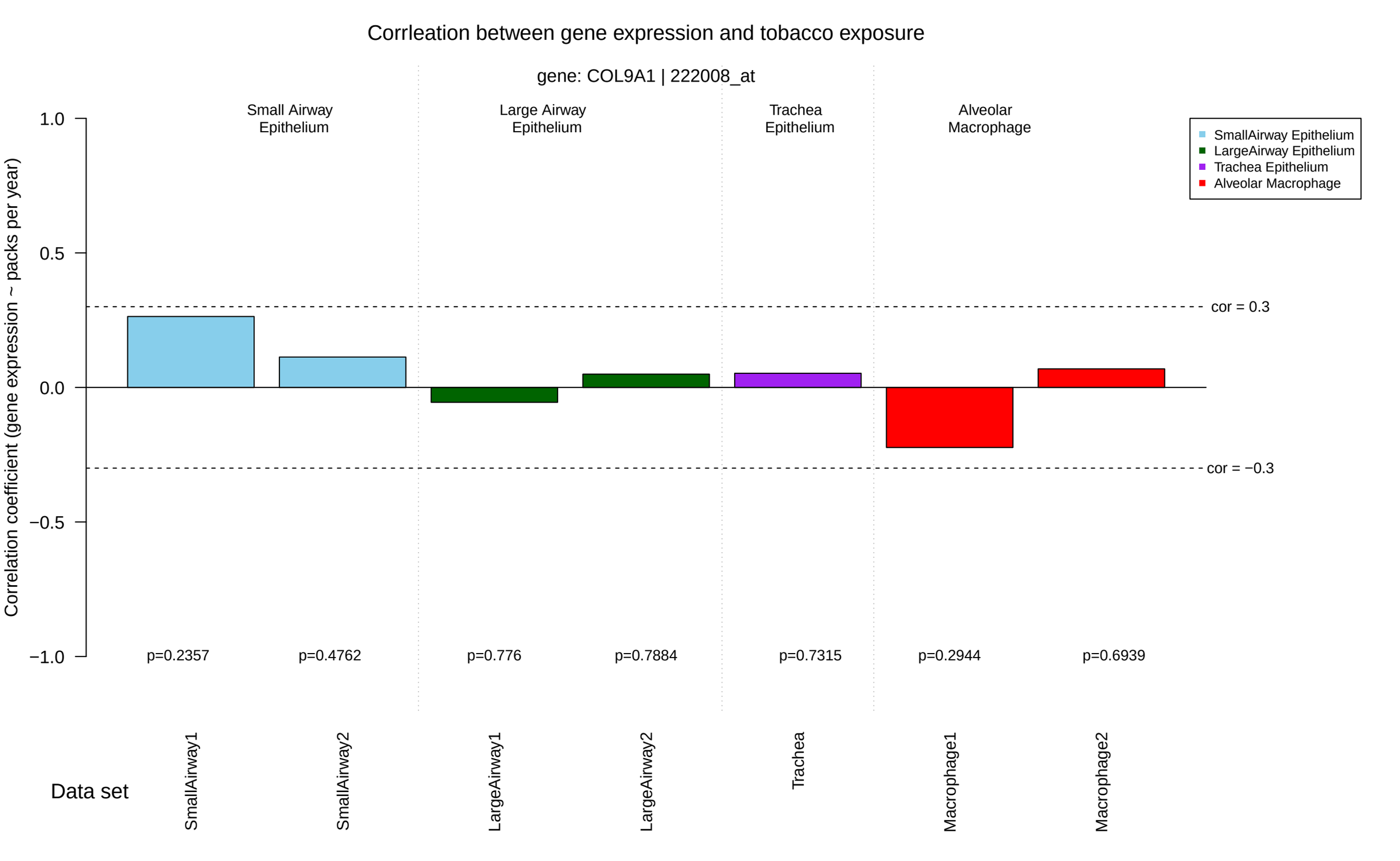

set: LargeAirway1 (p=0.6131)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1992)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2059)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9838)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.728)
COL9A1 | 222008_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL9A1 | 222008_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2357
p=0.4762
p=0.776
p=0.7884
p=0.7315
p=0.2944
p=0.6939
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2